ISSN 1943-7544

Pacific Asia Journal of the Association for Information Systems

Research Paper doi: 10.17205/1pais.23566 Volume 32, Issue 7 (2024)

Towards Federated Privacy-Preserving Machine Learning for Ad Targeting Across Heterogeneous Platforms

Arun Prasad, Grace Chen, Mohamed El-Sayed, Olena Vasylenko, Javier Martinez

[1*]Harbin Engineering University, China, 778523826@qq.com
[2]Beijing University of Technology, China, 2343423@bjut.edu.cn
[3]Harbin Engineering University, China, {hq234234, xxu14232, zl23422} @bjut.edu.cn

Abstract

This research investigates the potential of federated learning frameworks to enhance privacy in advertising targeting across diverse digital platforms and cultures. Leveraging secure aggregation protocols, differential privacy techniques, and multi-modal data fusion, the study demonstrates that PPML can facilitate collaborative ad model training while preserving consumer data confidentiality. Simulation results show robust performance, scalability, and legal compliance for multinational corporations, alongside challenges in fairness, personalization, and global deployment.

Keywords: Health Industry, Technology Convergence, Smart Medical, Technology Convergence Trend.

Introduction

The convergence of technology within the health industry has emerged as a transformative trend, reshaping the landscape of healthcare delivery. In the health industry, characterized by cross-border integration, an extensive industrial chain, and broad sectoral coverage (Jiao, 2020; Song, 2020; Sun, 2020), this trend spans medical and health services, scientific research, medicine, equipment, insurance, elderly care, food, education, tourism, and sports. The rapid development of convergent technologies has the potential to revolutionize these sectors and enhance healthcare outcomes. Despite its promise, the health industry currently faces significant technological bottlenecks. For example, hospitals face numerous challenges and issues related to storing, sharing and distributing patient health records using manual methods such as pen, paper and human memory. A technological environment that enhances efficiency, ensures cost-effectiveness, and facilitates timely decisions is urgently needed (Masrom & Rahimly, 2015). The increasing demand for high-quality medical services, driven by aging populations and rising income levels, also fuels the need for convergent technologies (Shagerdi et al., 2022; Won et al., 2016; Zhang et al., 2024). Technology offers immense potential to address these challenges, particularly by supporting independent living for the elderly, compensating for physical decline, and assisting with daily activities. Information technology opens up new possibilities through the application of various mobile

solutions in healthcare. For example, mobile text reminders can significantly improve medication adherence, follow-up rates and disease monitoring, as well as self-monitoring in long-term conditions such as chronic obstructive pulmonary disease (COPD), hypertension, and heart failure. These preventive measures can lead to significant reductions in hospitalizations(Yunara et al., 2023). These examples highlight the urgency of leveraging convergent technologies to address persistent challenges and innovate healthcare delivery systems.

Multiple factors, including market demand, economic growth, policy support, and advancements in technology, converge to drive the inevitable trend of technological integration in the health industry. First of all, market demand is influenced by three primary factors: the increase of the elderly population (Xing & Zhang, 2020; Zhou & Wang, 2019), the prevalence of chronic diseases (Amendola et al., 2014; Zheng & Rodríguez-Monroy, 2015), and the public's pursuit of a healthy life (Shi, 2016). By integrating various scientific and technological disciplines, health industry providers can develop new solutions that address evolving market needs and enhance patient outcomes. Secondly, the health industry has emerged as a significant economic growth driver. The convergence of big data , the Internet, and technological innovation in medical services has led to the development of intelligent healthcare, smart pensions, and other emerging sectors that fuse technology with health, creating new growth points for economic development (Choi et al., 2015; Li & Chang, 2020; Shao et al., 2015; Zhu & Wu, 2020). Thirdly, policy support at the national level, with "health" being a strategic priority in China (Cao et al., 2019) , further accelerates technological convergence within the health industry. This policy support provides a conducive environment for innovation and the adoption of new technologies. Finally, the advancement of Industry 4.0 and the core technology of the Fourth Industrial Revolution, such as artificial intelligence, have brought significant technological changes to traditional industries. The application of artificial intelligence in the health industry will inevitably provide innovative technical means for health governance (Wang, 2019). Advances in biological discovery, computational power, and biocompatible materials are key drivers of convergence in healthcare. These advancements enable the development of new medical technologies and services that can significantly improve healthcare delivery (Eselius et al., 2008; Sharp et al., 2016). These factors collectively drive deeper and more complex technological convergence within the health industry.

However, the increasing complexity of technological convergence also brings challenges. Despite growing academic interest, existing research on technological convergence in the health industry remains fragmented. Integrating diverse technologies across fragmented value networks introduces uncertainties and regulatory obstacles that must be managed carefully (Phillips et al., 2017). The main technologies driving convergence and the specific innovation directions they enable are not well-documented. Among them, the main technologies of technology convergence are those that enable technological convergence within the health industry, acting as key enablers of innovation and advancement. A comprehensive synthesis of evidence and insights into these aspects is essential for advancing the field.

This scoping review aims to distinguish itself from previous studies in several ways. Firstly, it adopts a more systematic and comprehensive approach to identify and map the existing evidence on technological convergence within the health industry. Unlike previous fragmented studies, our research aims to provide a holistic view and clear trends. Secondly, it focuses specifically on answering two research questions:

RQ 1: Which technological convergences have brought about innovations in the health industry?
RQ 2: What are the directions of innovation in the health industry driven by technology convergence?

Scoping reviews are particularly suited for synthesizing evidence in emerging fields where the literature is diverse and complex (Peters et al., 2020; Tricco et al., 2016). Arksey and O'Malley (2005) first defined the concept, describing scoping studies as a systematic form of knowledge synthesis that maps evidence to answer research questions and effectively collates research achievements to share with policymakers and practitioners. They are particularly useful for mapping existing knowledge, identifying gaps, and informing future research (Logan et al., 2024; Munn et al., 2018), especially in the health and social science disciplines(Pham et al., 2014; Tricco et al., 2016). Unlike theory-testing reviews such as qualitative systematic reviews and meta-analyses, scoping reviews—designed to map key concepts, technologies, and research gaps through broad literature overviews—do not necessitate formal quality appraisal. This makes scoping reviews particularly useful for synthesizing evidence in emerging or complex fields, where detailed quality assessments of studies may not yet be feasible (Levac et al., 2010). Given the exploratory nature of this study, a scoping review offers a robust framework for examining diverse research topics across multiple levels of evidence.

This study adopts a scoping review framework to analyze the development characteristics of the health industry, the main technologies driving convergence, and the directions of innovation, following the framework of " identifying research questions -- positioning and selection -- existing results analysis -- trend analysis". Data sources include Web of Science (WOS) and China National Knowledge Infrastructure (CNKI, https://cn.oversea.cnki.net/, a private-owned publishing company in China since 1999, operating databases of academic journals, conference proceedings, newspapers, reference works, and patents). The findings will provide evidence-based insights to guide future research, inform policymakers, and facilitate technological innovation in the health industry.

The remainder of this paper is organized as follows. The next section characterizes the health industry and summarizes existing literature on technological convergence. This is followed by a description of the study's methodology and data sources. Subsequently, the findings are presented and discussed. Finally, the paper concludes with a summary of insights and limitations.

Literature Review

This literature review explores the technological convergence in the health industry. It identifies gaps in existing research and emphasizes trend analysis as a key tool for understanding technological advancements and guiding future innovation.

Health Industry Related Work

The Connotation, Characteristics and Classification of the Health Industry

Sorting and analysis of the health industry from the connotation, health characteristics, and classification of the health industry:

The connotation of the health industry. From the overall and macro perspective, the health industry, as a burgeoning sector with immense potential, encompasses various activities aimed at maintaining, restoring, and promoting health. These activities include the production of health products, provision of service, and dissemination of health-related information (Zhang, 2018). From the perspective of industrial objectives, the health industry is tasked with continuously enhancing the overall level of human health and catering to the diverse needs of society for a healthy life. It operates through the close collaboration among multiple industries or companies directly involved in health -related endeavors, functioning as an organized entity or a nexus of economic activities wherein individuals offer health products or services (Kuo et al., 2020; Song, 2020; Zheng et al., 2019). From the perspective of technological innovation, scholars have directed their attention towards examining the evolution of the health industry following the integration of emerging technologies. For instance, the emergence of the Internet of Health Things (IoHT) has revolutionized healthcare by combining mobile applications, wearable devices and other interconnected devices for monitoring and managing health-related data (Masmali & Miah, 2021; Terry, 2016). Similarly, the concept of the Internet of Medical Things (IoMT) entails the integration of implantable and wearable devices, synchronized with personal smart devices like smartphones or smartwatches, and interconnected through the Internet (Jha, 2017). Moreover, healthCare4. 0 framework emphasizes the pivotal role of Information and Communication Technology (ICT) as the cornerstone of healthcare advancement. The continuous improvement in wireless technology has significantly facilitated real-time monitoring of physiological parameters, with the Internet of Things playing a vital role in medical, diagnostic and imaging sensor equipment (Aceto et al., 2020).

Characteristics of the health industry. The health industry is an intensive domain requiring significant knowledge, capital, human resources, and other inputs. It stands out from other industries due to its unique features, including broad coverage, an extended industrial chain, a high technical threshold, pronounced development through agglomeration, and substantial interdisciplinary complexity (Jiao, 2020; Song, 2020; Sun, 2020). Academician Zhong Nanshan emphasizes that the essence of the health industry lies in shifting the focus from treatment to prevention. He advocates transitioning from a treatment-centric approach to one centered on disease prevention. The health industry should evolve from merely addressing downstream demands to fostering upstream demand. This means empowering the public with the ability to prevent and detect health risks

early, thereby identifying and mitigating potential diseases at their inception. To achieve this, it is essential to implement measures that promote the practical application of preventive medicine, rehabilitation medicine and geriatric medicine. The widespread application of information technologies such as the internet and cloud computing has facilitated cross-industry convergence, which has become a significant trend in the current development of the health industry (Jiao, 2020).

Health industry classification. The connotation of the health industry has evolved. It is no longer limited to disease treatment alone but is expanding to encompass broader aspects, including medicine, medical treatment, health care, health management, disease prevention, medical care, etc. With the expansion of the scope of the health industry, the classification of the health industry has become more detailed and comprehensive. According to different classification standards, the health industry can be categorized into various segments. Firstly, based on the national economic industry classification, the health industry can be divided into 13 industrial categories (National Bureau of Statistics of China, 2019). Secondly, based on the nature of market demand, the health industry can be categorized into four segments: mature industries with rigid demand, emerging industries with rigid demand, mature industries with elastic demand, and emerging industries with elastic demand (Song, 2020). Finally, according to the different modes of service provision, the health industry can be divided into 5 major industry clusters: the medical industry, which focuses on medical service organizations; the pharmaceutical industry, which includes pharmaceuticals, medical devices and medical consumables; the health care industry, which revolves around health food and health products; the health management service industry, which encompasses health testing and assessment, consulting services, conditioning and rehabilitation, and protection and promotion; and the health care industry, characterized by convergence and cross-border cooperation (Zhang, 2020). Among them, the combination of medical care and elderly care transcends mere superposition; it emphasizes "convergence and cross-industry", integrating medical care, rehabilitation, health care, and health preservation through the convergence of new technologies such as big data, artificial intelligence, and the Internet of Things. This integration aims to deliver medical services, rehabilitation nursing, and health management to better meet medical needs (Ouyang, 2020). However, other divisions also exist, for example, the United States classifies the health industry into six categories: medicine, medical equipment, medical insurance, health management, medical care and healthy food (Ding, 2018).

It is evident that the health industry is trending towards interdisciplinary, cross-border convergence and technological innovation: Firstly, the connotation of the health industry extends beyond treatment, with increasing demand for prevention and health management broadening its scope. Secondly, from the perspective of the characteristics of the health industry, the industry's high technical threshold and significant interdisciplinary complexity are among its most vital and promising characteristics. Finally, from the perspective of the classification of the health industry, the "cross-border convergence-oriented medical care industry" has emerged as a major branch within the health sector. Therefore, future convergence and innovation are expected to drive technological advancements and transformative changes in the health industry.

The definition and description of these concepts and characteristics lay the foundation for the subsequent discussion, including the introduction of innovative technologies and the process of technological convergence in the health industry's management field. This article combines the macro significance of the health industry with the perspective of technological innovation, focusing on the activities of product production, service provision, and information dissemination aimed at maintaining, repairing, and promoting health following the introduction of emerging technologies. Based on this concept we develop a compendium of technology convergence in the health industry.

Research Related to Technology Convergence in the Health Industry

In the context of the new era, the health industry has undergone a significant transformation in its connotation. The integration of advanced technologies (such as artificial intelligence, the Internet of Things, etc.) has revolutionized of the traditional health industry. This transformation has led to the emergence of new medical diagnostic methods, including "digital health (Bhavnani et al., 2016) " and "smart medical care" (Fang et al., 2019; Qiu et al., 2015). The cross-penetration of technologies from other fields provides unprecedented opportunities and impetus for the development of the health industry.
Scholars have demonstrated the changes that various technologies on the health industry. The convergence of Internet of Things technology and related innovations in the health industry serves as a primary driving force for distributed healthcare applications. This convergence not only fosters the development of the healthcare sector but also contributes to the reduction of medical expenses (Couturier et al., 2012). Chen et al. identified the

Internet of Things, cloud computing and big data technology as the foundational pillars of the modern healthcare (Chen et al., 2013). Typical applications such as wearable devices provide essential methods and tools for personal health management (Fernandez & Pallis, 2014). Cloud computing meets the IT needs of the healthcare sector and further simplifies the medical process (Ahuja et al., 2012; Calabrese & Cannataro, 2015). Electronic health records (EHR), electronic medical records (EMR), and personal health records (PHR) can record clinicians' medical examinations, medical images, and treatment processes, thereby assisting clinicians in decision-making (Garets & Davis, 2006; Smolij & Dun, 2006).

The convergence of artificial intelligence (AI) technology and related innovations in the health industry significantly enhances diagnosis accuracy and reducing human error (Leone et al., 2021; Meskó et al., 2018). In addition, medical imaging and auxiliary diagnosis have achieved substantial breakthroughs with the aid of new technologies. Leveraging the development advantages of big data in medical imaging and image recognition technology, AI medical imaging has emerged as the most advanced field of artificial intelligence and medical convergence in China. The AI medical imaging industry has developed relatively mature products in areas such as pulmonary nodules, fundus, breast cancer, and cervical cancer. Medical imaging, electronic medical records, diagnosis robots, and virtual assistants are the primary application scenarios at the auxiliary diagnosis level, with products predominantly comprising integrated software and hardware solutions.

The convergence of robotics (edge computing technology) and related technologies in the health industry offers significant advantages in interactivity (friendly user interface), insight, navigation, and innovation (Wan et al., 2020). Typical applications include mobile medical service robots and social robots. In the field of health industry, mobile medical robots have several advantages over humans: (1) The learning ability of robots far exceeds that of humans, and can provide more effective diagnosis and treatment. (2) Medical robots have stronger adherence, and psychotherapeutic robots can be desktop devices connected to a smartphone (Jeong & Breazeal, 2017), which making it convenient to capture environmental details and patient psychological characteristics. (3) Robots excel at repetitive tasks and have a lower error rate. For example, robots can remind caregivers to arrange medication at specified time (Tao et al., 2016), the health industry has a low fault tolerance rate, where mistakes can have life-threatening consequences.

It is evident that technology convergence has already occurred in the health industry. The concept of technology convergence can be traced back to Rosenberg's definition in 1963, which describes it as a phenomenon where technological innovations across different industries merge to overcome various technical obstacles in the production processes (Rosenberg, 1976). Technology convergence involves the blurring of boundaries between distinct fields of science, technology, markets, and industries, focusing on the process of merging these fields (Curran & Leker, 2011; Lee, 2007). Starting from this concept, technological convergence is not an isolated process. Although driven by various factors, it also impacts societal and economic operations. Therefore, research on technological convergence should analyze it within the broader context in which it occurs (Chen et al., 2013). Hacklin proposed an evolutionary framework to analyze technology convergence. He divided the entire convergence process into four phases: knowledge convergence, technology convergence, application convergence and industry convergence (Hacklin, 2007). Curran and Leker posited that convergence was a comprehensive process encompassing science convergence, technology convergence, market convergence to industry convergence, as illustrated in Figure 1 (Curran & Leker, 2011). The process begins with the convergence of science, marked by the increasing overlap of different scientific fields and the strengthening of interdisciplinary research between coexisting industries. The second stage is technological convergence, characterized by the increasing overlap of different technological fields, leading to the emergence of new technological platforms. Thirdly, technological convergence may lead to market convergence, characterized by the combination of new products and markets, such as in technology-driven industries like electronics or chemicals. However, market convergence can also occur independently of scientific convergence steps, directly triggered by changes in customer needs (Schmidt et al., 2016). Finally, industry convergence completes the overall process, driven by the impact of technological or market convergence on the capabilities and competitiveness of companies, resulting in the emergence of new industries and sectors.

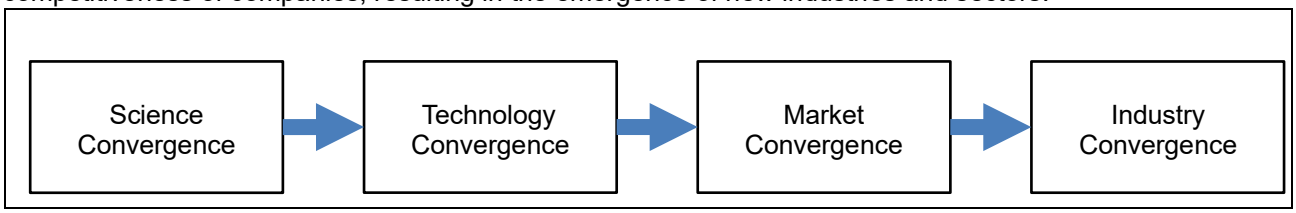

Figure 1 – Sequential Process of Convergence

The stage of research and analysis in this paper is currently in the first phase, focusing on analyzing knowledge flows in the literature data in order to pave the way for driving subsequent technology convergence. Technology convergence can foster the development and innovation of the health industry, enabling product innovation and iteration. However, the insights of technology convergence from individual studies on technology convergence in the health industry are often limited to addressing specific issues. To gain a more comprehensive perspective, it is essential to apply systematic analytical approaches, such as trend analysis, to evaluate the broader trajectory of technological advancements and their influence on the industry.

Trend Analysis Related Research

Trend analysis, a widely adopted methodology in social sciences, traces its origins back to the realms of statistics and economics. As early as the 19th century, statisticians embarked on utilizing time-series data to decipher the evolution of economic and societal phenomena (Mann, 1945). With the advancements in information science and computer technology, trend analysis has progressively permeated into a broader spectrum of disciplines, encompassing but not limited to social sciences, medicine, engineering, and information science. At its core, trend analysis lies in identifying and interpreting patterns of change within data over time, employing quantitative methods to unveil underlying developmental trends and periodic variations (Hyndman, 2018).

The pivotal concept of trend analysis revolves around recognizing and elucidating long-term patterns and trends embedded within time-series data. This process encompasses data collection, visualization, pattern recognition, and forecasting. By discerning long-term upward or downward trends, cyclical fluctuations, and anomalies within data, trend analysis empowers researchers to comprehend past performances, anticipate future changes, and inform strategic decision-making (Montgomery et al., 2015).

Within bibliometrics, trend analysis primarily serves to evaluate the knowledge development trajectory of a particular research topic or field, pinpointing research hotspots and future research directions (Smith et al., 2014). By scrutinizing metrics such as publication counts, citation patterns, and keyword frequencies, researchers gain insights into the evolution of disciplinary landscapes, thereby furnishing data-driven support for research policy formulation, academic evaluation, and research direction planning (Garfield, 1955).

Trend analysis assumes a crucial role in literature reviews. It not only facilitates researchers in systematically reviewing and summarizing the research history of a given field but also reveals the inherent laws and future trends of disciplinary development (Small & Griffith, 1974). Recent research underscores the significant cooccurrence between the concepts of "literature review," "bibliometrics," and "trend," exemplified by studies such as those by Amarathunga (2024), Gani et al. (2025), H. Xu et al. (2024), Liu et al. (2022), and Mishra et al. (2024). These studies harness bibliometrics as a research tool, utilizing methods like analysis, organization, classification, and clustering of literature to deduce future research trends, technological trends, or market trends. Furthermore, some literature employs sophisticated techniques like machine learning to enhance the analysis and clustering of literature.

Why is trend analysis so important? There are four primary reasons: 1) Revealing Long-Term Patterns: Trend analysis empowers researchers to uncover long-term patterns and trends within data, thereby identifying potential economic or market shifts (Pindyck, 2018). 2) Facilitating Interdisciplinary Research: Trend analysis can identify intersections between different disciplinary areas, fostering the initiation of interdisciplinary research endeavors (Gani et al., 2025). By bridging gaps between fields, it promotes innovation and the development of novel perspectives. 3) Forecasting Future Trends: Leveraging trend analysis, researchers can make predictions about future changes, aiding in the formulation of long-term plans and strategies (H. Xu et al., 2024; Pajalic et al., 2024). 4) Optimizing Resource Allocation: By pinpointing areas with growing demand or declining interest, trend analysis aids in optimizing resource allocation. This enables a more efficient utilization of resources, ensuring that funds, personnel, and other assets are directed towards areas of greatest potential impact (Pajalic et al., 2024).

Commentary

It is evident to find that the health industry is experiencing a trend of interdisciplinary, cross-border convergence and technological innovation: Firstly, the scope of the health industry extends beyond treatment alone, the increasing demand for prevention and health management has expanded its boundaries. Secondly, from the perspective of the characteristics of the health industry, the high technical threshold and the significant degree of interdisciplinary complexity are key positive aspects of the health industry. Finally, from the perspective of the classification of the health industry, the 'convergence of cross-border-oriented medical care' has emerged as a major branch within the health industry.
As evident from the numerous scholars who have studied technology in the health industry, new technologies such as the Internet of Things technology, cloud computing technology, and big data technology have innovated various service methods, operation methods and medical methods of the health industry. However, current research exhibits two main deficiencies: On the one hand, while most literature acknowledges the value of technology convergence for the health industry, these studies only analyze the impact of technology convergence on the health industry from a specific technological aspect. This results in fragmented studies that lack a systematic approach to understanding technology convergence in the health industry. Therefore, it is necessary to comprehensively organize and integrate the overall state and technological structure of technology convergence development in the health industry. This will aid in constructing an integrated framework for the development of technology convergence in the health industry. On the other hand, the predominant technology for technology convergence in the health industry remains unclear. What types of technology are required to support the health industry? What innovations will be driven by technology convergence in the health sector? The absence of such information hinders the ability of health enterprises to overarching the general trend of technological development, the evolutionary trajectory of the health industry, and the formulation of competitive strategies. Furthermore, it impedes evidence-based decision-making of policymakers. Hence, to ascertain the extent and trajectory of technological convergence in the health industry, this paper will conduct a scoping review to delineate the converging technologies in the health industry. Among these, trend analysis serves as a strategic decision-making support, a potent tool for enhancing performance, and bolstering adaptability to ever-evolving conditions. This critical analytical approach forms a significant direction of this paper, offering recommendations for promoting technological innovation within the healthcare industry and guiding enterprises in their technological adoption.

Method

In order to comprehensively analyze and understand the current research status of the technology convergence in the health industry, this study utilized data from two major databases: WOS and CNKI. Web of Science (WOS) is a multidisciplinary database that publishes high-quality, peer-reviewed research across various fields, including health and medical sciences, which is crucial for this review. Meanwhile, CNKI, as one of the largest Chinese academic resource platforms, provides extensive access to localized literature that complements the international coverage of WOS. Its inclusion ensures that this review captures valuable insights into research trends in China and the broader Asia-Pacific region, offering a balanced perspective on global and regional developments in the health sector.

This study adopted a structured review framework informed by the systematic review methodology of Denyer and Tranfield (2009). While the original five-step framework—"search, location, research selection and evaluation, analysis and synthesis, reporting results"—provides a robust foundation, this review refines and adapts it into a four-step process tailored to the context of technology convergence in the health industry: “1) Identifying research questions; 2) Positioning and selection; 3) Analysis of existing results; 4) Trend analysis", the specific process is as Figure 2. This refined framework ensures a systematic and transparent approach to analyzing the interplay between technological advancements and innovation directions in the health industry. By combining international and localized literature, the review provides a comprehensive understanding of the subject matter while addressing existing gaps in the literature.

Identify the Research Questions

To comprehensively understand the development of technology convergence in the health industry, this paper unifies the literature with similar themes, re-analyzes, compares, and interprets findings and evidence from

existing studies. By condensing the analogies that have emerged in different studies, it enhances the dialogue and accumulation of knowledge within the field. Furthermore, it identifies previously undiscovered implicit phenomena from the literature, thereby enriching the current knowledge system with new insights. Developing research questions and objectives is crucial for a scoping review to address broad research questions and map evidence from various sources(Pollock et al., 2021). The information ultimately obtained in this paper should contribute to the development of an integrated framework for technology convergence in the health industry, supporting health firms in driving technological innovation. This article focuses on analyzing the main converging technologies in the health industry and the directions in industrial innovation. Therefore, the research questions of this article are centered around the following: *RQ 1: Which technological convergences have brought about innovations in the health industry?*
RQ 2: What are the directions of innovation in the health industry driven by technology convergence?

Positioning and Selection

Positioning: The Search Strategies

To comprehensively collect literature data on technology convergence in the health industry, this study adopted a tailored search strategy for two major databases: CNKI and WOS. Each database has unique characteristics that require specific approaches to ensure the inclusion of diverse and relevant literature (Booth et al., 2012; Bramer et al., 2017).

CNKI, as the largest Chinese academic database, reflects the localized research landscape and policy-oriented development of China's health industry. Its unified conceptual framework, namely the "Classification of Health Industry Statistics" by the National Bureau of Statistics, ensures consistency in terminology and definitions (National Bureau of Statistics, 2019). In contrast, WOS provides access to a broader, global perspective, accommodating diverse terminologies used by scholars worldwide. This distinction necessitated the development of database-specific search strategies to capture both international advancements and localized characteristics in the health industry.

A "pearl-growing" approach (Booth et al., 2012) was employed to iteratively refine the scope of keywords. "*Pearlgrowing is an alternative way to start developing your search strategy. Pearl-growing refers to identifying a known highly relevant article (the 'pearl') to identify terms on which a search can subsequently be based. If you haven't already identified a key article for your topic area, you can conduct a very specific search, such as looking for key concepts in an article title to identify a key reference*". Starting with " health industry" as a core theme, preliminary searches identified key terminologies and concepts in the literature, which were expanded upon to include technical and thematic variations across both databases.

Search Strategy for the CNKI: The CNKI search strategy was guided by the 2019 official definition and classification of the health industry, which encompasses medical and health care, biotechnology, prevent diseases, and life sciences. Specifically, the National Bureau of Statistics clearly defined the health industry as "a collection of production activities that provide products (goods and services) directly or closely related to health for the public, based on medical and health care, biotechnology, and life sciences, with the purpose of maintaining, improving, and promoting the health of the people". Therefore, keyword "health industry" inherently cover core concepts like biotechnology and chronic disease/ disease prevention. To address variations in terminology related to technology convergence, expressions such as "*("Technology Convergence" OR "Technology Cross-border" OR "Convergence Technology" OR "Converging Technology" OR "Technology fusion")*" were included, ensuring comprehensive retrieval of Chinese literature.

Search Strategy for the WOS: The WOS strategy balanced alignment with CNKI keywords and the broader international research landscape. Definitions from scholars such as (Donzé & Fernández Pérez, 2019; Göran & Whitehead, 1991; Relman, 1991; Schweitzer & Lu, 2018; Yamamoto et al., 2023) and institutions like the WHO (World Health Organization, 2016, 2018) and others (Company, 2021; PwC, 2021) informed the inclusion of terms like "healthcare," "medical," " chronic disease/diseases prevention, " "biotechnology," and "life sciences." In addition, the introduction of the concept of "chronic disease/disease prevention" into the retrieval strategy is due, on the one hand, to the fact that this concept has been included in the scope of the health industry by numerous scholars (Göran & Whitehead, 1991; World Health Organization, 2016, 2018), and on the other hand, to ensuring consistency with the retrieval scope of CNKI. To capture studies on technology convergence, terms

like “technology convergence” and its variations were included. Non-technical keywords, such as "insurance," were excluded to maintain focus on technology-related studies.

To reconcile differences in scope and terminology between CNKI and WOS, overlapping keywords were prioritized, while unique terms were cross-validated against the conceptual definitions provided by authoritative sources. This approach ensured consistency in capturing the breadth of technology convergence research while addressing the nuances of each database. To enhance the quality and relevance of the dataset, during the retrieval process, the literature was restricted to refereed journal articles, excluding book chapters, conference papers, and reports. Such restrictions ensured the inclusion of high-impact studies representing the forefront of research (Wibowo et al., 2022) in technology convergence and health industry innovation. The detailed strategies are presented in Table 1.

Table 1 – Search Strategies for WOS and CNKI

Database	Search strategy
WOS	TS=(((healthcare*) or (health*) or (healthy*) or (medical*) or (nurse*) or (nursing*) or (care*) or pharmaceuticals or biotechnology or ("life sciences") or ("chronic disease") or disease) and (("Technology convergence") or ("Technology fusion") or ("converging Technology")))
CNKI	Theme = (("Health industry") AND ("Technology Convergence" OR "Technology Crossborder" OR "Convergence Technology" OR "Converging Technology" OR "Technology fusion"))

Selection: The Screening Criteria

We adhere the operational steps of scoping review analysis, involving the extracting, labelling and categorizing the literature data. After retrieving the data from WOS and CNKI, data screening process needs to be conducted. The data screening process is synthesized and compared from the studies of Ansari and Gupta (2021), Kumar and Gupta (2023), Mitchell and El-Gayar (2022) and Perdana and Kulkarni (2023). Considering the volume of data in this study, the data selection process is defined as: Invalid data and duplicates deletion—Records screened by full text—Forward & Backward Search. Here, invalid data refers to cases where data cannot be obtained or abstracts are missing. After removing the invalid data and duplicates, and applying the inclusion and exclusion criteria outlined in Table 2, the dataset is refined. To further enhance the comprehensiveness and robustness of the dataset, the Forward & Backward Search strategy is employed. This iterative process helps to uncover potentially overlooked but significant contributions, thereby enriching the overall dataset and strengthening the foundation for subsequent analysis (Kumar & Gupta, 2023). Articles identified through targeted searches were included or excluded following a thorough reading and review of each paper. During the screening process, all three authors independently assessed each article based on the criteria outlined in Table 2. Any discrepancies in judgment were discussed and resolved through consensus.

- The focus is on market convergence, industry convergence, or scientific convergence.
- Attention is given to protein fusion and organismal fusion. Not contribute to the answers to the research questions (RQ1 and RQ2) of this review. The excluded literature is unable to answer any of the research questions.

Data Coding

To systematically analyze the literature on technology convergence in the health industry, this study employed a combination of structured and open coding methods. The structured coding focused on predefined categories based on the research questions, while the open coding allowed for the identification of emergent themes not captured by the initial framework.

Structured Framework

The initial coding framework was informed by the research questions and prior literature. It incorporated the following main categories: (1) Technology Types: Identified based on the research objects specified by the authors, focusing on mutually converged technologies mentioned in the literature, such as artificial intelligence,

biotechnology, and Internet of Things (IoT). (2) Innovation Directions: Extracting health industry domains where these technologies are applied, and highlighting specific outcomes or trends resulting from technological convergence, such as precision medicine, personalized treatments, and cross-industry integration. These categories guided the initial analysis and ensured consistency across the dataset.

In our results, the terms IT, ICT, computing technologies, and communication technologies frequently appeared. To avoid conceptual overlap, we referred to the literature for a brief distinction among these terms: IT focuses on information management (e.g., M. Xu et al. (2024)); ICT integrates information and communication for data exchange (e.g., Eidam et al. (2017)); computing technologies handle data processing and analysis (e.g., Chen et al. (2020)); and communication technologies facilitate the transmission of information in the health industry (e.g., Yun & Geum (2019)).

Open Coding Process

In addition to structured coding, open coding was applied to identify novel concepts and patterns beyond the predefined framework. The process involved the following steps:

Initial Reading: Each article was read in detail, and emergent keywords, phrases, concepts or relevant findings contributing to the answers for the research questions were noted. It is worth mentioning that in this context, reading the full text of the articles is crucial, rather than limiting oneself to abstracts and titles. The complexity of comprehending scientific papers often leads scholars to only rely on the title, abstract, and keywords, overlooking the valuable information embedded in the body of a paper. However, the real technological potential is demonstrated and discussed in the internal sections of a paper, making it imperative for scholars to delve deeper into scientific content rather than resorting to superficial analysis (Cammarano et al., 2024). Indeed, the primary source for obtaining research conclusions lies in the discussion, analysis, and conclusion sections, which often provide detailed insights into the analysis of technological convergence, as well as a thorough explanation of the occurrence process of technological convergence, the main technologies involved, and the innovative directions. Meanwhile, it should be clear that there may be one or more main technologies and innovation directions in an article. For instance, unexpected terms such as "health tourism" or "blockchainenabled health management" were identified during this phase.

Iterative refinement: The open coding process was repeated iteratively until no new themes emerged, ensuring the saturation of categories.

Thematic Clustering: Innovation directions were clustered into broader themes based on function and objectives. For example, mentions of "prevention of illness," "health management," and "healthy aging" were clustered under the theme " prevention and health management." Ultimately, five main groups emerged: Prevention and Health Management, which focuses on maintaining and promoting health (AbdulRaheem, 2023; Jain et al., 2018; Mehanna, 2022; Sehrawat, 2023); Intelligentization of Medical Technology, emphasizing the role of intelligent technologies in enhancing the accessibility, efficiency, and precision of healthcare services (Adibi et al., 2024); Precision and Personalized Medicine, aiming to develop customized medical solutions tailored to individual differences, aligning with advancements in precision medicine (Elemento, 2020; Raskina et al., 2017); Pharmaceutical Innovation, pertaining to research and development advancements in the pharmaceutical sector (Haider, 2023); and Integration of Medical Services and Industries, capturing the crosssector convergence of healthcare with other industries.

Cross-validation and Consistency

To ensure the reliability of the coding process, two researchers (XG and HM) independently conducted open coding on the articles. Discrepancies were resolved through discussion with the third researcher (LC), and a Cohen's Kappa coefficient of 0.83 indicated strong inter-coder agreement.

Results

Overview of Included Studies

This approach identified 88 articles, including results until December 2024. Firstly, after removing the invalid data and duplicates, 82 articles left. Secondly, applying the inclusion and exclusion criteria in table 2. The screening of records by full text involves manually reviewing the full text of the literature to determine whether they are relevant to the technology convergence of health industry, and contribute to the answers to the research questions. The process can be further refined into three steps: (1) Determine whether the research topic falls within the scope of the health industry. Although some articles may be retrieved based on search strategies, their research topics may deviate from the health industry (for example, discussions on the ethical issues of emerging technologies) or may constitute noise information (e.g., health weather index). (2) Assess whether the research topic focuses on technological convergence. Some articles may mention technological convergence, but their primary focus may be on the convergence of scientific knowledge, market integration, or industrial convergence. (3) Determine whether the article provides answers to the research questions posed in this review, specifically whether it discusses the key technologies in health industry technological convergence (for RQ1), or/and whether it explores the innovative directions of technological convergence in the health industry (for RQ2). That results in 41 articles left. Finally, the Forward & Backward Search strategy is employed, leading to the inclusion of the full texts of 55 articles in the final review. The detailed process is illustrated in Figure 3. The complete list of literature is provided in Appendix A.

Technology Convergence Structure and Overall Trend

To provide a more intuitive representation of technology convergence, we used the Gephi social network analysis tool to visualize the technology co-occurrence network (Figure 4). The visualization highlights that the health industry is currently centered around key technologies such as artificial intelligence, big data, information technology, and biotechnology, which serve as foundational pillars for technological advancements. Additionally, the industry is actively integrating emerging technologies like machine learning, sensing technology, nanotechnology, cloud computing, 5G, blockchain, and the metaverse. The interconnectedness of these technologies underscores the growing trend of convergence, which is expected to drive continuous innovation and enhance the efficiency, accessibility, and precision of healthcare services.

Information Technology

Statistical analysis identified information technology as the leading technology subject of convergence in the health industry, with a total of 15 related articles. The convergence of information technology can be primarily divided into two directions: 1) Convergence with nanotechnology and biotechnology for pharmaceutical innovation. The convergence of biotechnology, nanotechnology, and information technology is expected to establish precision biology and personalized medicine (Lee, 2010). Maine et al. (2014) explored the process of technological convergence in therapeutic nanoparticle drug development through two case studies of pharmaceutical innovation. Similarly, Du et al. (2019) explored the innovation mechanism of medicine innovation, suggesting that such innovation arises from the convergence of clinical medicine, pharmacology, chemistry, biotechnology, immunology, and other related disciplines. They emphasized that successful innovation in the health industry requires a substantial accumulation of knowledge and a strong foundational base. 2) The convergence of information technology with other related digital technologies in the health sector, such as internet technologies (Tufano & Karras, 2005), communication technologies (Tufano & Karras, 2005), digital twin technology (Gliszczyński & Ciszewska-Mlinarič, 2021), sensor technology (Kang, 2020), and the Internet of Things (IoT) (Cao et al., 2019), plays a key role in optimizing healthcare services, innovating health management, and improving resource allocation efficiency. Through the deep convergence of information technology with other digital technologies, the health industry can more effectively meet the diverse needs for health and medical care, driving the high-quality development of healthcare. For example, products such as wearable devices, portable health monitoring equipment, self-service health testing devices, smart elderly care monitoring systems, and home service robots are already in use on the market, offering solutions for chronic

disease management, home-based health and elderly care, and personalized health management (Wu & Tan, 2019).

Artificial Intelligence

The second is artificial intelligence (AI). The co-occurrence network analysis reveals that AI occupies the central position, establishing connections with numerous other technologies. The convergence structure is divided into two major directions: digital technology and biotechnology. The convergence with digital technology primarily aids the health industry in medical diagnosis and detection, enhances the efficiency and quality of medical services, promotes the intelligent transformation of the medical industry, and addresses social needs and innovation drives. The literature covers AI and its branch technologies, for example, image analysis is widely used in medical imaging, such as CT scans, tomography, and X-rays (Lee et al., 2021). Seales (2021) highlighted the critical role of AI in surgical procedures. The convergence of AI and related technologies has driven the development of smart healthcare (Wang & Lee, 2023). Medical testing technologies require AI to foster innovation (Cheng et al., 2022). In addition, the convergence with biotechnology mainly improves drug research and development and production optimization, as well as biomedical devices in the health industry. For example, in the pharmaceutical field, advances in machine learning not only enhance supply chain efficiency by simplifying manufacturing processes but also accelerate drug discovery through the use of large compound libraries (Aaldering & Song, 2021). The implementation of AI technology in biomedical devices, such as biosensors, has facilitated the intelligent monitoring of diseases and the recording of vital signs. This advancement has broadened the application scope of biotechnology within the healthcare sector, enabling biotechnology to contribute to disease diagnosis and health management with greater precision (Lim et al., 2021). The introduction of AI is of significant importance for the reform and innovation of the health industry. With AI support, the health sector can achieve efficient and intelligent upgrades in diagnosis, health management, resource allocation, and drug development, while greatly improving the accessibility and quality of medical services. This, in turn, will enhance global health levels and promote the continued development of the health industry.

Biotechnology

Thirdly, biotechnology, as a pivotal technology in the health industry, boasts extensive applications encompassing precision medicine, gene editing, molecular diagnostics, vaccine development, and more. It serves as a fundamental cornerstone for the innovative development of the health industry. There are 13 relevant articles, and key application methods can be summarized into the following three categories: Firstly, precision medicine and personalized healthcare. As mentioned previously, the convergence of biotechnology with information technology represents a significant convergence trend. Precision medicine is a new medical field pioneered by genomic technology and bioinformatics. Secondly, drug research and development, as well as manufacturing. Biotechnology, combined with nanotechnology, has achieved breakthroughs in the field of drug delivery, exemplified by the development of targeted nano-therapeutics such as Doxil® and radiolabeled antibody drugs like Zevalin® (Maine et al., 2014). Notably, Aaldering and Song (2021) research indicates that the trend of biotechnology for pharmaceutical innovation has been waning, with emerging trends suggesting that AI and big data are playing an increasingly significant role in the pharmaceutical sector. Thirdly, medical technologies and diagnostic tools. Advances in biotechnology have propelled the application of molecular biology and genetic engineering, enabling the development of novel diagnostic tools (such as electrode-based sensing chips for RNA virus detection) to enhance diagnostic accuracy and efficiency (Zhu & Xie, 2022). The convergence of biotechnology with 3D printing technology can reshape medical cells and organs, opening up new possibilities for regenerative medicine (Zhao & Zhang, 2019).

Big Data Technology

The fourth is big data technology, whose integration often manifests in the convergence with other digital technologies, such as artificial intelligence (Aaldering & Song, 2021; Gao & Jiang, 2024), the Internet of Things (Fan et al., 2021), cloud computing (Bhatt et al., 2021; Fan et al., 2021), and blockchain (Bhatt et al., 2021; Fan et al., 2021). The application of big data technology in the health industry has increasingly become a pivotal force in driving industry innovation and optimizing medical services. It plays a crucial role in the collection, storage, processing, and analysis of medical data, significantly enhancing the efficiency, accuracy, and personalization of medical services. The convergence of big data technology in the health industry primarily addresses three key issues: data collection and integration, innovative types of health service methods (Han, 2018; Yang & Hou, 2021) and contributing to precision medicine and personalized services (Gao & Jiang, 2024;

M. Xu et al., 2024). Utilizing extensive databases, this approach integrates resources to develop new technologies, production processes, and business models within the industry. The platform's communication capabilities are leveraged to facilitate technology research and development, alongside the trial of new products in the medical and health professional fields (He et al., 2020; Song et al., 2021). Additionally, big data technology is employed to develop innovative health service methods, such as smart medical care. This includes the research and development of reliable smart health monitoring equipment and other mobile products, the creation of highly personalized medical records, and the deep convergence of big data and information technology. By linking personal health information with hospitals, doctors, and medical equipment, doctors can comprehensively understand an individual's health status over time. This not only saves time and medical expenses but also enhances medical convenience (Ma, 2021; Song et al., 2021).

Internet Technology

Internet technology also plays a pivotal role, which frequently integrates with various other digital technologies, such as communication technology (Tufano & Karras, 2005; Wu & Tan, 2019) and information technology (He et al., 2020; Song et al., 2021). The application of the Internet in the health industry has facilitated a comprehensive digital transformation, ranging from medical services to health management. Its deep convergence with technologies such as big data, artificial intelligence, the Internet of Things, and blockchain has not only optimized existing medical service models but also spurred the emergence of new industry forms and business models. Firstly, it provides educational and training support. Leveraging Internet resources, crossborder training can be conducted, such as Internet-based lecture-style training modes, which help enhance the training standards of global medical and health-related professionals, provide health assistance to developing countries, and facilitate the dissemination and improvement of medical knowledge and skills (Jiao, 2020; Solez et al., 2005). Secondly, the health industry utilizes Internet technology as a medium to gradually integrate into other industries (Song & An, 2020). For example, companies leverage information technology platforms
(including mobile Internet, big data, smart terminal equipment) platforms to innovate and optimize the production and operation of sports service products suitable for the elderly. This convergence shortens the distance between supply and demand, increases the frequency of interaction, enhances user trust, and completes the realization loop, thereby achieving a model of integrated development through cross-penetration between smart sports and smart elderly care services. The "Vintage Fitness" studio in Canada provides door-to-door or remote private fitness guidance, health knowledge lectures, group course training, and daily care services (Han, 2018). This service model has been expanded through the Internet to enhance the traditional Chinese medicine (TCM) service model by integrating it with modern medical practices. The focus is not only on the treatment of diseases but also on diseases prevention, daily health maintenance, and post-illness rehabilitation. This approach aims to establish a comprehensive, standardized and professional TCM physical examination and prevention system. It also seeks to expand the scale of health services and rehabilitation departments while promoting and popularizing TCM knowledge (Song et al., 2021). Thirdly, optimizing medical processes and innovating service models (Song et al., 2021; Wu & Tan, 2019). Internet technology has achieved process optimization in the medical industry through informatization means, including electronic health records (EHRs), teleconsultation, and remote imaging analysis. These technologies not only improve the efficiency and accuracy of medical services but also drive innovation in medical service models, particularly the emergence of the "Internet + Healthcare" model. Through the integration of Internet platforms, various medical services (such as medication purchases, health consultations, and online consultations) can be seamlessly connected, greatly enhancing patients' medical experience.

Other Technologies

In addition to the aforementioned five technologies, other significant technologies such as sensing technology (Kang, 2020; Wang & Lee, 2023) and the Internet of Things (IoT) (Fan et al., 2021; Lim et al., 2021) have facilitated the expansion of health management from medical institutions to households and individuals. Nanotechnology (Shim et al., 2016; Song et al., 2017), blockchain (Bhatt et al., 2021; Fan et al., 2021), virtual reality (Wang & Hsu, 2023), and 3D printing (Lim et al., 2021) have brought groundbreaking innovations to the health industry. Communication technology (Yun & Geum, 2019), cloud computing (Bhatt et al., 2021), and 5G (Ma, 2021) have empowered the remotization and digitization of medical services.

Overall, the current technological focus of the health industry revolves around the convergence of information technology and biotechnology, while emphasizing data processing and analysis, the application of internet and

communication technologies, sensing and monitoring technologies, and the exploration and application of emerging technologies. The aim is to enhance the medical service level, innovation capability, and data management capability of the health industry.

The Directions of the Health Industry Innovation Driven by Technology Convergence

The literature was sorted and labeled to analyze the innovation directions of health industry driven by technology convergence, aiming to ascertain functional realization. The statistical analysis of the innovation directions is presented in the Table 4. Through reading and analysis of the literature, and combining the technology fusion and innovation directions mentioned in the conclusions, we have formed a total of 20 labels. Among them, the five directions with the highest frequency in the literature are prevention of illness, personalized healthcare, intelligent medical technology, telemedicine, and biopharmaceuticals. Subsequently, these 20 innovation directions were summarized and grouped into five categories. Below, we will discuss and analyze these five innovation directions.

Innovation directions	**Labeled keywords**	**Σ**	
		WOS	**CNKI**
Prevention and Health Management (29)	**Prevention of Illness**	6	5
	Health Management	2	5
	Smart Healthcare	3	4
	Healthy Aging	3	0
	Smart Aging	2	0
Intelligentization of Medical Technology (27)	**Intelligent Medical Technology**	8	1
	Telemedicine	8	0
	Mobile Healthcare	1	3
	Remote Monitoring	0	1
	Medical Informatization	2	1
	Informatization of Traditional Chinese Medicine	0	2
Precision Medicine and Personalized Medicine (17)	**Personalized Healthcare**	8	2
	Precision Healthcare	0	1
	Precision Medicine	2	4
Biopharmaceuticals and Pharmaceutical Innovation (11)	**Biopharmaceuticals**	8	0
	Pharmaceutical Innovation	3	0
Integration of Medical Services and Industries (6)	Integration of Healthcare and Elderly Care	0	2
	Medical Tourism	0	2
	Internet-based Healthcare	0	1
	Medical Consultation	0	1

Prevention and Health Management

An analysis of 55 articles revealed that 29 focused on prevention and health management, highlighting its paramount importance, with more than 50% of the relevant research recognizing its significance. As the most widely applied direction in the integration of health industry technologies, it emphasizes disease prevention and health management, particularly in individual health monitoring, early diagnosis, and lifestyle interventions. Technologies such as intelligent health and smart elderly care will increasingly focus on disease prevention, health maintenance, and healthy aging management. For instance, the combination of biosensors and AI technologies can monitor various diseases and record vital signs, aiding in early detection and prevention of diseases (Afifuddin & Seo, 2024). This also includes the dissemination and learning of concepts, such as

enhancing disease prevention and control management through the promotion of traditional nursing methods (Wu & Tan, 2019).

Looking ahead, the future health industry will shift from a singular treatment model to one emphasizing disease prevention and health management. Health management systems will integrate functions such as remote monitoring, personalized health plans, and preventive healthcare, enabling individuals to manage their health and prevent diseases from the source, thereby achieving proactive health management. This transformation aligns with the development trend of modern health concepts and will effectively improve the overall health of the population and reduce the burden of disease.

Intelligentization of Medical Technology

A total of 27 articles addressed intelligent medical technology, accounting for nearly 50% of the literature. Undoubtedly, the intelligentization of medical technology is one of the core trends in future healthcare services, encompassing areas such as telemedicine, mobile healthcare, and intelligent medical technologies. Remote monitoring (Meng et al., 2013), medical informatization (Gliszczyński & Ciszewska-Mlinarič, 2021; Song et al., 2017), and the digitalization of traditional Chinese medicine (Cao et al., 2019) will make healthcare services more efficient, convenient, and precise.

With the development of intelligent medical technologies, telemedicine and mobile healthcare are expected to become more widespread. Particularly in remote areas and among aging populations, telemedicine will offer more accessible healthcare services and reduce the imbalance in medical resource distribution. Hospitals and clinics will also enhance service efficiency through medical informatization, utilizing data sharing, electronic health records, and intelligent diagnostic systems to accelerate medical decision-making and treatment processes, thus significantly improving the quality of care.

Precision Medicine and Personalized Healthcare

A total of 17 articles addressed precision medicine and personalized healthcare. Influenced by the highfrequency terms such as personalized healthcare, precision healthcare, and precision medicine, this area is expected to become a core development direction in the future health industry. Through advanced technologies such as genomics (Aaldering & Song, 2021), big data analytics (Yun & Geum, 2019), and artificial intelligence (Cheng et al., 2022), healthcare can provide highly personalized and tailored medical treatments for patients. This innovative approach applies not only to disease treatment but also to disease prevention and health management. The development of precision and personalized medicine will revolutionize traditional healthcare models, making services more aligned with individual needs, improving treatment outcomes, reducing medical resource waste, and driving the health industry towards more refined and precise practices.

Biopharmaceuticals and Pharmaceutical Innovation

A total of 11 articles addressed biopharmaceuticals and pharmaceutical innovation. Biopharmaceuticals and drug innovation will continue to introduce new medications to meet the treatment needs of complex diseases, playing a crucial role in precision and personalized medicine (Lee, 2010). With advancements in biotechnology and gene editing, the pharmaceutical industry will develop more customized drugs, such as precision medicines for complex diseases like cancer and autoimmune disorders (Jiang & Luan, 2018). Pharmaceutical innovation will improve treatment efficiency through targeted drug delivery and molecular-targeted therapies (Maine et al., 2014). New treatment methods, such as personalized medicines and cell therapies, will offer more precise treatment options. The continued advancement of biopharmaceuticals and pharmaceutical innovation will provide powerful tools for combating major diseases and drive further breakthroughs in drug development within the health industry.

Integration of Medical Services and Industries

A total of six relevant literatures addressed the integration of healthcare services and industry. With the advent of an aging society, the integration of medical and elderly care services (medical-elderly care integration) has emerged as a significant trend. Additionally, medical tourism is poised to become an emerging industry, attracting global patients through the combination of tourism and medical services (Song et al., 2021; Zhao & Zhang, 2019), particularly in the fields of rehabilitation and recuperation, and medical aesthetics. Furthermore, the

development of telemedicine and online medical consultation services has facilitated the global sharing and popularization of medical resources, enabling patients to access quality medical services beyond geographical constraints. The integration of medical services and industries will expand the boundaries of the health industry, create more new business forms and models, and inject new vitality into the sustainable development of the health industry.

In summary, the future health industry will thrive around five innovative directions: prevention-oriented care, precision medicine, intellectualization, pharmaceutical innovation, and cross-industry integration. Preventive medicine, personalized treatment, and health management will constitute the mainstay of industry development. Innovations in biopharmaceuticals and medical technologies will lead the industry towards greater efficiency and precision. With the deepening development of an aging society, emerging models such as medical-elderly care integration and medical tourism will become important growth points. The widespread application of telemedicine and digital health technologies will promote the popularization and convenience of global medical services, making greater contributions to the health and well-being of people worldwide.

Discussion

This study aimed to systematically explore technological convergence and its impact on innovation within the health industry. By addressing two research questions, we identified the main technological domains driving innovation and outlined the primary directions of innovation enabled by such convergence.

Key Findings

This scoping review aimed to map the literature on technology convergence and innovation facing the health industry, 55 papers were published on the subject. The main technologies driving innovation and outlined emerging innovation trends are assessed. This scoping review synthesizes 55 studies, highlighting the transformative role of technological convergence in reshaping the health industry. The findings reveal that technological convergence fosters innovation by connecting previously isolated fields, offering pathways for enhanced healthcare delivery. These technological convergences not only transform the service models, operating methods, and medical practices of the health industry but also drive its sustained development towards prevention-oriented approaches, precision medicine, intelligentization, pharmaceutical innovation, and cross-industry integration.

To address the first research question, in terms of the key technological aspects of technological convergence in the health industry, this study clarifies the core status and integration structure of multiple key technological fields. Specifically, the study identifies information technology, artificial intelligence, biotechnology, big data, and internet technology as pivotal to technological convergence. Information technology, as the primary technological domain, exhibits two major directions of convergence: with nanotechnology and biotechnology for pharmaceutical innovation, and with other digital technologies to optimize medical services. These findings confirm the foundational role of IT in data management and connectivity (Tufano & Karras, 2005). Artificial intelligence technology occupies a central position in the co-occurrence network, with its branch technologies applied in various aspects such as medical image analysis, pharmaceuticals, surgery, smart health, and medical testing, which highlighting the evolving prominence of AI in applications such as diagnostics and personalized medicine (Lim et al., 2021). Biotechnology, as a core technology in the health industry, has widespread applications in precision medicine, drug discovery and preparation, medical technologies, and diagnostic tools. However, studies indicate that its trend in pharmaceutical innovation has weakened, while the influence of AI and big data has increased, reflecting the dynamic process of technological convergence in the health industry and consistent with current trends in digitally-driven pharmaceutical research and development (Aaldering & Song, 2021). These results challenge earlier perspectives that emphasized biotechnology as the primary driver of innovation, instead positioning digital technologies as key enablers of convergence. By expanding the understanding of integration mechanisms, this study provides theoretical insights into the evolving roles of core technologies in industrial innovation. Big data technology focuses on exploring demand for health products and developing new types of health services. Internet technology enriches the dissemination paths and communication modes of the health industry, promoting industrial penetration and business model innovation, which echoes the general theory of the Internet in facilitating industrial integration and innovation (Solez et al., 2005). In addition to the aforementioned technologies, sensing technology and the Internet of Things (IoT) drive the expansion of health management, while the metaverse, blockchain, virtual reality, and 3D printing bring

breakthrough innovations. Communication technology, cloud computing, and 5G empower the remotization and digitization of medical services.

To address the second research question, our findings delineate five innovation directions: prevention and health management, intelligentization of medical technology, precision medicine, biopharmaceutical advancements, and cross-industry integration. For instance, the emphasis on health management aligns with the theoretical shift towards prevention-oriented healthcare systems (Wu & Tan, 2019). The convergence of telemedicine and mobile health technologies breaks geographical barriers, enhancing access to medical services—a critical aspect in aging societies (Gliszczyński & Ciszewska-Mlinarič, 2021). Additionally, the fusion of medical and elderly care services underscores the potential for cross-industry synergies, contributing to the literature on industry convergence and diversification.
These innovation directions are closely interconnected, forming an intrinsic relationship. Prevention and health management provide the foundation and prerequisites for other innovation directions. Early health interventions and management can reduce the incidence of diseases, thereby alleviating the pressure on intelligent medical technologies, precision medicine, and personalized medicine, while also establishing a broader base of healthy populations for integrated models such as the combination of medical care and elderly care. The intelligentization of medical technologies offers the technical means for precision and personalized medicine, enabling the precise collection and analysis of medical data through smart devices and digital platforms, which in turn supports the formulation of tailored treatment plans. The advancement of precision and personalized medicine further drives biopharmaceutical and pharmaceutical innovation, encouraging pharmaceutical companies to develop drugs and treatment methods that are better tailored to individual differences. Finally, the convergence of medical services with other industries provides broader development opportunities for the health industry as a whole, fostering the organic combination of health with related sectors, enabling resource sharing, and promoting coordinated development.

Theoretical Implications

The health industry is in a developmental stage, and there is a lack of comprehensive understanding regarding the current status of research within the health industry. Utilizing a scoping review methodology, this paper constructs a theoretical framework coupling multi-source data. This framework follows a structured approach, including "identification of research questions - positioning and selection - analysis of existing results - trend analysis". It is characterized by its universality and objectivity, providing a robust foundation for the synthesis of research findings. Specifically, theoretical contributions can be divided into three aspects: First, the review framework developed in this study integrates previous research efforts, addressing the fragmented state of research on technology convergence in the health industry. Through the scoping review and analysis process, this paper assimilates insights from various perspectives, particularly focusing on the main technologies driving health industry convergence and the innovation directions within the industry. Therefore, the refinement of health industry technology convergence further enriches and advances the body of research related to the health industry. Secondly, this study supports the conceptual framework of convergence as a driver of crossdisciplinary innovation, advancing theories on technological convergence and industry transformation (Curran & Leker, 2011). Thirdly, the analysis provides empirical support for the growing importance of information and communication technology (ICT) in enabling systemic change within traditional industries, contributing to the broader literature on Industry 4.0.

Practical Implications

Through the review of technology convergence in the health industry and the analysis of emerging trends, this study can provide practical insights from two primary perspectives. On the one hand, by identifying the current research focuses and deficiencies, it offers new research perspectives and directions for technological innovation for researchers. On the other hand, it can provide strategic insights and support for the product innovation and development directions of health enterprises, enhancing their overall competitiveness of health enterprises in China and other Asia-Pacific regions. The ability of health enterprises to dynamically integrate and reconfigure knowledge is paramount to sustaining a competitive edge, as highlighted by Björkdahl (2009). Technological convergence precedes market convergence, which in turn forms the foundation for industrial penetration. Notably, the initial step towards transcending industrial boundaries necessitates breaking down knowledge convergence barriers, as depicted in Figure 1 (Curran & Leker, 2011). Drawing upon literature-based data (at the knowledge level), this paper forecasts technological convergence within the health industry. Within the process of technological convergence in the health industry, some enterprises emerge as pioneers, while others follow suit. This dichotomy is crucial in determining whether health enterprises can shape their competitiveness amidst ever-changing environments. Therefore, this study provides strong support for the

progress of the global health industry and also offers new starting points for subsequent research in optimizing technological convergence, interdisciplinary collaboration, and addressing ethical and legal challenges.

Limitations

This paper acknowledges several limitations. Firstly, in terms of data, the search results predominantly focus on the combination of material technology within the health industry while neglecting the spiritual needs of the elderly. Secondly, the study overlooks variations in research subjects, the differences of the research object are ignored, such as the disparities between urban and rural areas and regional differences. Henceforth, future research endeavors could be refined in two primary directions: 1) Revision of Search Strategies: It is imperative to revise the search strategies to augment the depth and breadth of investigations, acknowledging that addressing the spiritual needs of the elderly constitutes an integral aspect of the health industry. For instance, incorporating specific keywords such as "companion robots," "chatbots," and related terms into the data retrieval strategies can target the provision of companionship, conversation, and interaction, thereby mitigating loneliness and fulfilling the spiritual requirements of the elderly. This approach ensures a more comprehensive exploration of technologies aimed at enhancing their well-being. 2) Undertaking Categorical Studies: During the data acquisition and processing phases, it is advisable to embark on segmented research, particularly focusing on distinguishing between urban and rural elderly populations. By conducting comparative analyses of the similarities and differences in main technologies and innovation directions across these regions, we can tailor policies and corporate strategies more effectively based on their unique needs. This categorization not only facilitates a nuanced understanding of the challenges faced by each segment but also informs targeted interventions and product development, ensuring greater relevance and impact.

Conclusion

This paper presents a comprehensive scoping review of technological convergence in the health industry, synthesizing fragmented research findings to provide a systematic understanding of the current state of convergence within this sector. By integrating insights from existing studies, this research identifies main technologies and industrial innovation directions, constructing a structural framework that captures the mechanisms and impacts of technological convergence. The findings not only elucidate the technological and industrial landscape of the health industry but also highlight critical trends. Looking forward, technological convergence is poised to play a transformative role in shaping the health industry's response to global demographic and economic shifts. In particular, the convergence of mobile and digital solutions has significant potential to meet the needs of aging populations, enabling independent living and enhancing the quality of life for the elderly. Future research should focus on exploring the ethical and regulatory dimensions of such convergence, ensuring that innovation is both inclusive and sustainable. By advancing our understanding of technological convergence, this study lays the groundwork for future innovations that will drive the evolution of the global health industry, especially the Asia-Pacific region.

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